

Databasutveckling och lagring — Final Exam: RETAKE II

2025-03-06

Instructions

1. No computers, no phones, no notes
2. No talking with other people in the room
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Points* box!)
5. For each question, you'll need to answer the question by crossing with an **X** the right answer (multiple choice questions) or write your answer in the empty lines below the question (open questions).
6. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Point system

There are **10 questions**, worth **2 points** each, for a total of maximum **20 points**.

There are no partial points

You either give the **complete, correct answer**, and get 2 points, or get 0 points.

Grading

Points	Grade
0 - 12	IG
13 - 19	G
20	VG

READ EVERYTHING CAREFULLY!

1 Normalization

Given the following schema of a SQLite database:

Listing 1: The complete schema for the DB

```
PRAGMA foreign_keys = ON;

CREATE TABLE IF NOT EXISTS customer (
  customer_id INTEGER PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  postal_code VARCHAR(5) NOT NULL,
  city VARCHAR(50) NOT NULL
);

CREATE TABLE IF NOT EXISTS orders (
  order_id INTEGER PRIMARY KEY,
  order_date TEXT NOT NULL,
  customer_id INTEGER NOT NULL,
  FOREIGN KEY (customer_id) REFERENCES customer(customer_id)
);

CREATE TABLE IF NOT EXISTS products (
  product_id INTEGER PRIMARY KEY,
  product_name TEXT NOT NULL,
  category TEXT,
  unit_price FLOAT NOT NULL
);

CREATE TABLE IF NOT EXISTS order_items (
  order_id INTEGER NOT NULL,
  product_id INTEGER NOT NULL,
  quantity INTEGER NOT NULL,
  product_name TEXT NOT NULL,
  PRIMARY KEY (order_id, product_id),
  FOREIGN KEY (order_id) REFERENCES orders(order_id),
  FOREIGN KEY (product_id) REFERENCES products(product_id)
);
```

Point out and explain **one** normalization problem present in the schema.

 Order-items-tabellen innehåller en sammansatt PK men "quantity" beror bara på order-id, inte product-id. Så ej 2NF.

- 2 Explain how you would solve the normalization problem you described above.

You don't need to write a SQL query, just explain how you would fix it.

Jag hade lagt quantity i orders-tabellen.

- 3 In this course we've labbed with a database that contained the data for a music store. What was the name of the database, and which RDBMS did it use?

Name: Vi använde oss av den i botten av kursen

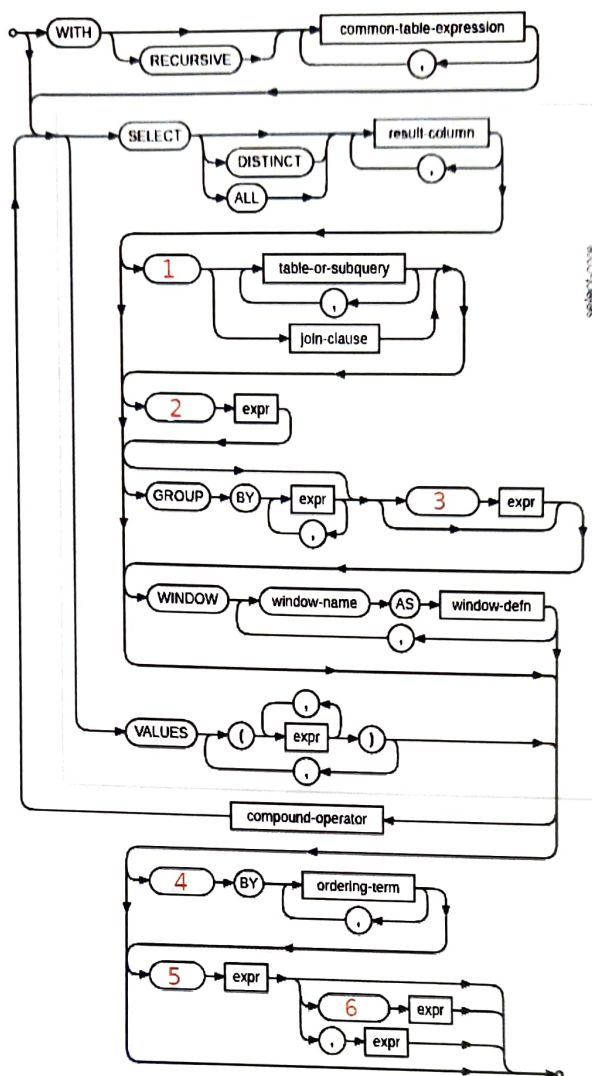
RDBMS: SQLite

- 4 In this course, we've labbed with a database that contained the data for a movie rental store. What was the name of the database, and which RDBMS did it use?

Name: Hanhka of data

RDBMS: SQLite

5 Fill the blanks



- 1: FROM
- 2: WHERE
- 3: _____
- 4: _____
- 5: _____
- 6: _____

6 What's the difference between WHERE and HAVING in a SELECT query?

- WHERE är ett villkor(?) tex där något ska uppfylla något tex WHERE a < b.
- HAVING är ett krav(?)

7 What's the difference between 'JOIN...USING' and 'JOIN...ON'?

Ingen?

8 We can always replace 'JOIN...ON' with 'JOIN...USING'

- ① True
- 2. False

9 We can always replace 'JOIN...USING' with 'JOIN...ON'

- ① True
- 2. False

10 What is a subquery?

Def as query i en query.